

A Fair Value Engine for Kalshi Sports Markets: Binary Option Pricing via Brownian Motion

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Abstract

Kalshi event contracts settle at \$1 if an event occurs and \$0 otherwise, which makes them mathematically identical to cash-or-nothing binary options. This document derives a fair value model for Kalshi “team X wins” game markets in two layers: a pre-game Elo-based prior, and an in-game model where the score margin is treated as a Brownian motion and the win probability follows directly from the distribution of its terminal value. We then use this fair value to define market edge and size positions via the Kelly criterion. Every result here follows from undergraduate probability: the CDF of the normal distribution, elementary properties of Brownian motion, and Kelly’s log-utility argument.

1 Kalshi contracts as binary options

A Kalshi “YES” contract on an event E pays

$$\text{Payoff} = \begin{cases} \$1 & \text{if } E \text{ occurs} \\ \$0 & \text{otherwise} \end{cases}$$

Under the assumption of no arbitrage and risk-neutral pricing, the fair price of this contract today is

$$V_0 = e^{-r\tau} \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_E] = e^{-r\tau} \mathbb{Q}(E),$$

where $\mathbb{Q}$ is the risk-neutral measure and τ is the (typically short, hours-to-days) time to settlement. Because τ is small for a single game, $e^{-r\tau} \approx 1$, so

$$V_0 \approx \mathbb{Q}(E).$$

This is the key fact that makes Kalshi different from a normal asset: the *price itself* is (approximately) a probability. A market quote of 63¢ is a direct claim: “the market believes there is a 63% chance of E .” Our entire project is about producing an independent estimate of $\mathbb{Q}(E)$ (call it p_{model}) and comparing it to the market’s implied estimate $p_{\text{market}} = V_0$.

2 Layer 1: pre-game probability via Elo ratings

Before a game starts, we need a prior belief about each team’s strength. The Elo system (originally designed for chess) gives each team i a rating $R_i \in \mathbb{R}$, and converts a rating difference into a win probability via the logistic function:

$$p_{\text{home}} = \frac{1}{1 + 10^{-(R_{\text{home}} + H - R_{\text{away}})/400}}$$

where H is a fixed home-field-advantage bonus (in rating points). After each game, ratings update toward the observed outcome:

$$R'_{\text{home}} = R_{\text{home}} + K(S_{\text{home}} - p_{\text{home}}), \quad S_{\text{home}} \in \{0, 1\}$$

This is a simple stochastic approximation / gradient update: it nudges the rating in the direction of the surprise $(S - p)$, scaled by learning rate K . Elo can be derived formally as an online logistic regression with a single feature (rating difference) updated by gradient descent on the log loss, which is worth knowing but not required to use it. In this project Elo supplies the **prior** at kickoff ($\tau = 1$ below); the in-game model in Section 3 is what does the interesting stochastic work.

3 Layer 2: in-game probability via Brownian motion (Stern 1994)

3.1 Model setup

Let $M(t)$ denote the point margin (home score minus away score) at time t during the game, and let $\tau(t) \in (0, 1]$ denote the *fraction of the game remaining* at time t (so $\tau = 1$ at kickoff and $\tau \rightarrow 0$ at the final whistle).

Assumption. The margin process behaves like a driftless Brownian motion in “fraction-of-game-remaining” time:

$$M(t) - M(t_0) \sim \mathcal{N}(0, \sigma^2(\tau_0 - \tau))$$

for $t > t_0$, where σ^2 is a fixed volatility parameter (units: points² per full game).

Why this is a reasonable approximation. Scoring in football and basketball happens through a sequence of discrete, roughly independent possessions, each contributing a small random point swing. By the functional Central Limit Theorem (Donsker’s theorem), a sum of many small, roughly independent, roughly identically distributed increments converges to a Brownian motion as the number of increments grows. A game has enough possessions (order of 10–20 in the NFL, far more in the NBA) that this approximation is reasonable, though it is not exact – see the limitations in Section 6.

3.2 Deriving the win probability formula

Fix the current time t_0 with current margin $m = M(t_0)$ and fraction of game remaining $\tau = \tau(t_0)$. The home team wins if and only if the *final* margin $M_{\text{final}} = M(1)$ (using $\tau = 0$ as the endpoint) is positive.

By the model assumption, conditional on the current state,

$$M_{\text{final}} \sim \mathcal{N}(m, \sigma^2\tau).$$

Proposition 1. *Under the above model,*

$$\mathbb{P}(\text{home wins} \mid m, \tau) = \Phi\left(\frac{m}{\sigma\sqrt{\tau}}\right)$$

where Φ is the standard normal CDF.

Proof. Standardize M_{final} : let $Z = \frac{M_{\text{final}} - m}{\sigma\sqrt{\tau}}$, so $Z \sim \mathcal{N}(0, 1)$. Then

$$\mathbb{P}(M_{\text{final}} > 0) = \mathbb{P}\left(Z > \frac{-m}{\sigma\sqrt{\tau}}\right) = 1 - \Phi\left(\frac{-m}{\sigma\sqrt{\tau}}\right) = \Phi\left(\frac{m}{\sigma\sqrt{\tau}}\right),$$

using the symmetry of the standard normal distribution, $1 - \Phi(-x) = \Phi(x)$. □

This is the entire model implemented in `stern_model.py`. Two sanity checks confirm it behaves correctly:

- As $\tau \rightarrow 0$ (end of game) with $m \neq 0$: $\frac{m}{\sigma\sqrt{\tau}} \rightarrow \pm\infty$, so the probability collapses to 0 or 1 – correct, since the outcome is essentially decided once the clock runs out.
- At $\tau = 1$ (kickoff) with $m = 0$: the probability is exactly $\Phi(0) = 0.5$ – correct, an untied prior with no score yet should be a coin flip *if we ignore the pre-game Elo information*. In practice we blend in the Elo prior most strongly when τ is close to 1, precisely because the pure score-based signal is uninformative there (see `stern_model.py`, `blend_weight` parameter).

3.3 Connection to first-passage / hitting-time problems

An equivalent way to view this result: the home team loses if and only if $M(t)$ hits the barrier at 0 from above before the clock expires (once trailing, ties are possible but let’s set aside the discreteness). This frames the win probability as a *hitting probability* for Brownian motion against a fixed barrier, a standard object in stochastic calculus that also underlies barrier option pricing in the Black-Scholes framework. The techniques are the same family used to price knock-in/knock-out options; this project is a discrete-outcome, sports-flavored version of the exact same mathematics.

4 Calibrating σ

σ is the only free parameter in the model, and it must be estimated from historical data before the model can be used. Rearranging the model assumption:

$$Y := M_{\text{final}} - M(t_0) \sim \mathcal{N}(0, \sigma^2 \tau) \implies \frac{Y}{\sqrt{\tau}} \sim \mathcal{N}(0, \sigma^2)$$

regardless of τ . So, pooling every (game, snapshot) pair across historical data, the sample variance of $Y/\sqrt{\tau}$ is a consistent estimator of σ^2 :

$$\hat{\sigma}^2 = \widehat{\text{Var}}\left(\frac{Y}{\sqrt{\tau}}\right) = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{y_i}{\sqrt{\tau_i}} - \bar{y}/\sqrt{\tau}\right)^2.$$

This is simple method-of-moments estimation (implemented in `fit_sigma.py`) – no maximum likelihood machinery is required because the model assumption directly gives us a quantity with constant variance to estimate.

5 Measuring model quality: calibration and the Brier score

A probabilistic forecast is **calibrated** if, among all the times the model says “probability p ”, the event actually happens a fraction p of the time. Formally, binning predictions into buckets B_k and checking

$$\frac{1}{|B_k|} \sum_{i \in B_k} \mathbf{1}[\text{home won}_i] \approx \bar{p}_{B_k}$$

for every bucket k is exactly the reliability diagram produced in `calibration.py`.

A complementary scalar summary is the **Brier score**, the mean squared error between predicted probability and the binary outcome:

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - o_i)^2, \quad o_i \in \{0, 1\}.$$

It is minimized (in expectation) exactly when p_i equals the true probability of the event, which is why it is a *proper scoring rule*: a forecaster cannot improve their expected score by reporting anything other than their honest belief. A constant 0.5 forecast scores 0.25; our backtest on simulated data (Section 7) scores well below that.

6 From edge to position size: the Kelly criterion

Suppose the model believes $\mathbb{P}(E) = p$, but the market prices a YES contract at q dollars (so a NO-side profit of q per contract if E does not occur, and a YES-side profit of $1 - q$ per contract if it does). If we stake a fraction f of our bankroll W on YES, our bankroll after the outcome is either

$$W(1 + f \cdot b) \quad (\text{win, probability } p), \quad W(1 - f) \quad (\text{lose, probability } 1 - p)$$

where $b = \frac{1 - q}{q}$ is the net odds per dollar staked. Kelly's criterion chooses f to maximize the expected log bankroll growth rate:

$$g(f) = p \ln(1 + fb) + (1 - p) \ln(1 - f).$$

Differentiating and setting $g'(f) = 0$:

$$g'(f) = \frac{pb}{1 + fb} - \frac{1 - p}{1 - f} = 0 \implies f^* = \frac{pb - (1 - p)}{b} = \frac{p(1 + b) - 1}{b}.$$

This is the formula implemented in `kelly.py`. Two practical notes:

- **Fractional Kelly.** f^* assumes p is known exactly. Since p is really p_{model} , a noisy estimate, we stake only a fraction (e.g. one quarter) of f^* to reduce the damage from model error – a standard, well-justified adjustment in practice.
- **Slippage.** The formula above uses the quoted price q , but real fills are worse than the quote for any nonzero size. `kelly.py` haircuts q by a fixed number of cents before computing f^* as a simple, conservative correction.

7 Empirical validation on simulated data

Before trusting this pipeline on real data, we validate it on data simulated directly from the model's own assumptions (implemented in `generate_mock_data.py`): margin paths are literally simulated as Brownian motion with a known $\sigma = 13.5$. This is a standard sanity check in quantitative work: if the estimation and backtesting code cannot recover known ground truth from data generated by the model itself, there is a bug, full stop.

Result. Fitting σ on 300 simulated games (4,500 snapshots) recovered $\hat{\sigma} = 13.54$, against a true value of 13.5. The resulting calibration plot (Figure 1) shows predicted and realized win rates

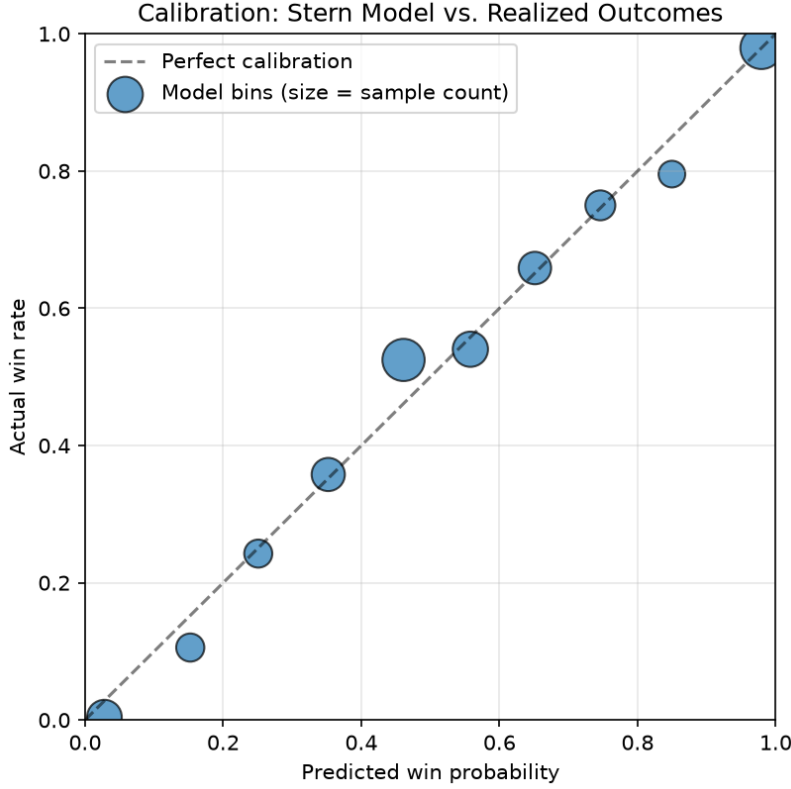


Figure 1: Reliability diagram on simulated data. Point size reflects the number of observations in each probability bucket.

closely tracking the 45-degree line across all probability buckets, and the backtest Brier score was 0.157, well below the 0.25 coin-flip baseline.

This confirms the pipeline is implemented correctly. The next step (see README) is to re-run the same pipeline on real historical play-by-play data, where $\hat{\sigma}$ and the calibration plot will reflect genuine model error rather than pure estimation noise – that gap between “clean-data calibration” and “real-data calibration” is itself an informative measurement of how good the Brownian-motion approximation actually is for real games.

8 Limitations and honest caveats

- **Constant σ .** Real scoring volatility is not constant across a game – end-of-game situations (intentional fouling in basketball, clock management in football) systematically distort the simple driftless-Brownian-motion picture. A natural extension is to fit $\sigma(\tau)$ as a function of time remaining rather than a single constant.
- **No drift term.** We assumed zero drift in the in-game model, folding all pre-game strength information into the Elo prior instead. An alternative model could carry a drift term through the whole game.
- **Discreteness.** Brownian motion is continuous, but football and basketball scores jump in increments of 2, 3, 6, 7, or 8 points. The approximation is reasonable away from the very end

of the game but breaks down near a tied score with little time left, where exact possession-based models (Markov chains over game states) do better.

- **Market microstructure.** We compare our model probability to Kalshi's last traded price, but real execution faces a bid-ask spread and thin order books; the Kelly sizing in `kelly.py` includes only a simple slippage haircut, not a full market-impact model.